

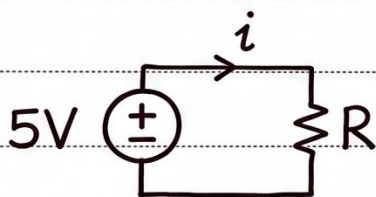
Why do we use imaginary numbers
in AC circuit analysis with capacitors
and Inductors?

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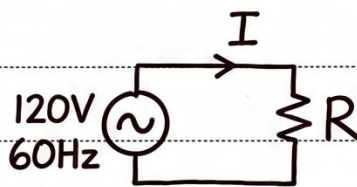
Why do we use imaginary numbers in AC Circuit analysis?

In DC, when having only a resistor: ratio between V and I stays the same at any time t ($R = V/I$), hence V and I are in phase. Same thing occurs in AC when we have only a resistor.



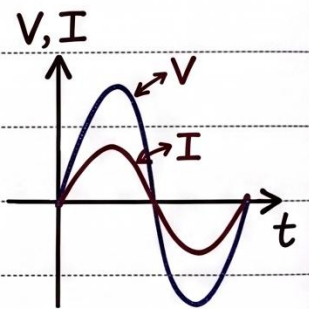
$$i = \frac{V}{R}$$

$$\phi = 0$$



$$I = \frac{V}{R}$$

$$\phi = 0$$



...With capacitors or inductors, this isn't the case...

$$i = C \frac{dv}{dt}$$

$$v = L \frac{di}{dt}$$

Clearly not in phase
when using sinusoidal

$$v = \sin t$$

functions (sin, cos)

$$C = 1 \Rightarrow i = \cos t = \sin(t + 90^\circ)$$

Now, we can apply KCL here, but that would be messy.

Applying KVL:

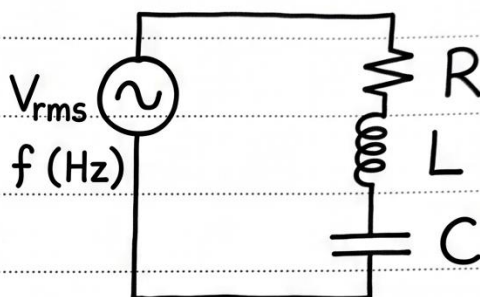
$$V_S = V_R + V_L + V_C$$

$$V_S = V_{rms} \cos(t)$$

$$V_R = V_{Rmax} \cos(t)$$

$$V_L = V_{Lmax} \cos(t + 90^\circ)$$

$$V_C = V_{Cmax} \cos(t - 90^\circ)$$



$$V_{rms} \cos(t) = V_{Rmax} \cos(t) + V_{Cmax} \cos(t - 90^\circ) + V_{Lmax} \cos(t + 90^\circ)$$

→ For a simple circuit, this equation is too difficult to deal with. You can use sine and cosine rules

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \sin \beta \cos \alpha$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

Now to find any unknown (like V_{Rmax}), you would require using trig identities and experience math trauma.

OR, you can use Euler's formula:

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

→ any trig expression can be rewritten as exponential.

$$\text{Also! see: } 1 + i = \sqrt{2} \cos(45^\circ) + i(\sqrt{2} \sin(45^\circ)) = \sqrt{2} e^{i(45^\circ)}$$

there is always a way to write complex number in this format.

~~For a capacitor. If we say $V = I/2\pi C$, but if we ignore phase for a moment and assume the equation:~~

Where the hell did this come from?

let's say Voltage across a component is $V_{\text{comp}} = A \cos(t + \phi_{\text{comp}})$, and we want to find the current across it. We can't just multiply V_{comp} by a constant to get the equation of I_{comp} sure multiplying by (-1) can phase shift by 180° , but we don't even know if I is shifted by 180° . So what do we do?

let's ^{observe} ~~see~~ this beauty for a second to see if it can benefit us:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$2e^{i30^\circ} = 2 \cos(30^\circ) + 2i \sin(30^\circ)$$

multiply both:

$$e^{i\theta} \times 2e^{i30^\circ} = 2e^{i\theta+i30^\circ} = 2e^{i(\theta+30^\circ)}$$

$$2e^{i(\theta+30^\circ)} = 2 \cos(\theta+30^\circ) + 2i \sin(\theta+30^\circ)$$

by multiplying two functions, we got a new function that has a magnitude of and phase shifted 30° .

Surely we can use this in our main problem of V_{comp} and I_{comp} to find I_{comp} across a capacitor or an inductor.

You can also use Euler's formula to find the math identity of any sinusoid function.

For example:

$$\cos(2\theta) = ?$$

we can just place that as the angle in Euler's formula

$$\Rightarrow e^{2\theta i} = \cos(2\theta) + i \sin(2\theta)$$

Hey! this is the same function we wanted to find its value!

How can we find it? we can evaluate $e^{2\theta i} \neq$ and find its real component, and that real component would equal $\cos(2\theta)$

$$\begin{aligned} e^{2\theta} &= e^{2(i\theta)} = (e^{i\theta})^2 = (\cos \theta + i \sin \theta)^2 \\ &= \cos^2 \theta + 2i \cos \theta \sin \theta - \sin^2 \theta \quad \leftarrow i^2 = -1 \\ &= \underbrace{\cos^2 \theta - \sin^2 \theta}_{\text{real component}} + \underbrace{i 2 \cos \theta \sin \theta}_{\text{imaginary component}} \end{aligned}$$

$$\text{So, } \cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

Also, if we set imaginary components equal to one another:

$$i \sin 2\theta = i 2 \sin \theta \cos \theta$$

Now let's apply that to the capacitor equation: $i = C \frac{du}{dt}$

note: to avoid confusion with current (i), imaginary symbol will be j . $j = \sqrt{-1}$

if $U = A \cos(\omega t + \phi)$, and $C = 1$ (for simplicity)
So, $i = -\omega C A \sin(\omega t + \phi)$

let's instead say:

$$U = A e^{j(\omega t + \phi)}$$

makes no sense, but the real component is our voltage

$$= A \cos(\omega t + \phi) + j A \sin(\omega t + \phi)$$

So, by the end we will see what the real component is

$$i = \frac{du}{dt} = \frac{d}{dt} (A e^{j(\omega t + \phi)}) = j\omega C A e^{j(\omega t + \phi)}$$

So, we can say, $i = j\omega C \cdot \bar{V}$ *

$$i = j\omega C \times (A \cos(\omega t + \phi) + j A \sin(\omega t + \phi))$$

$$= -\omega C A \sin(\omega t + \phi) + j\omega C A \cos(\omega t + \phi)$$

real component

is still i as we got normally by deriving $(\frac{du}{dt})$

but! It + gave us an equation of that is equivalent but doesn't involve differentiation.

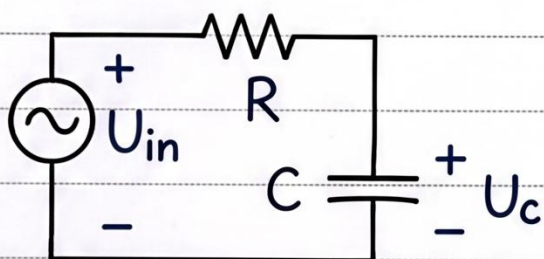
$$i = j\omega C \bar{V} \Leftrightarrow i = C \frac{du}{dt}$$

Now we can rearrange $i = j\omega CV$ to:

$$\frac{v}{i} = \frac{1}{j\omega C} = X_c \quad \text{which is just the impedance of the capacitor to current.}$$

Notice: $R = \frac{U}{i}$, and $X_c = \frac{U}{I}$ and the j of the imaginary numbers accounts for a phase shift of 90° just like $\cos$ and $\sin$.

Now let's put a real example:



applying KVL:

$$U_{in} = U_c + U_R$$

$$U_R = iR = C \cdot \frac{dU_c}{dt} \cdot R = RC \cdot \frac{dU_c}{dt}$$

$$U_{in} = U_c + RC \cdot \frac{dU_c}{dt}$$

we know U_c would be of the form $U_c = A \cos(\omega t + \theta)$, but instead of substituting that, which wouldn't be fun, we can use the earlier function we came up with.

$$U_c = A e^{j(\omega t + \theta)}, \text{ and } \frac{dU_c}{dt} = j\omega U_c$$

$$\text{So, } U_{in} = U_c + RC(j\omega U_c) = U_c(1 + j\omega RC)$$

and we finally get:

$$U_c = \frac{1}{1 + j\omega RC} \cdot U_{in}$$



cont.

For simplicity, $\omega = C = R = 1$

$$V_c = \frac{1}{1+j} V_i = \frac{1}{1+j} \times \frac{1-j}{1-j} V_i$$

$$= \frac{1-j}{1-j^2} V_i = \frac{1-j}{1+1} V_i = \frac{1}{2} V_i - \frac{j}{2} V_i$$

$$V_c = \frac{1}{2} V_i - \frac{j}{2} V_i$$

so, if $V_i = 3 \cos(t + 10^\circ)$

we can use euler's formula, $V_i = 3e^{j(t+10^\circ)}$

note: while putting in mind that the actual voltage of $V_i = 3e^{j(t+10^\circ)}$ is its real component.

$$V_c = \frac{\sqrt{2}}{2} e^{j(-45^\circ)} V_i$$

$$V_c = \frac{\sqrt{2}}{2} e^{j(-45^\circ)} * (3e^{j(t+10^\circ)})$$

$$V_c = \frac{3\sqrt{2}}{2} e^{j(t-35^\circ)}, \text{ but of course we only care about the real component.}$$

$$V_c = \frac{3\sqrt{2}}{2} \cos(t - 35^\circ)$$

↑
this result is how imaginary numbers help us.

For simplicity, $\omega = C = R = 1$

$$\begin{aligned} U_c &= \frac{1}{1+j} \cdot U_i = \frac{1}{1+j} \cdot \frac{1-j}{1-j} \cdot U_i \\ &= \frac{1-j}{1-j^2} \cdot U_i = \frac{1-j}{1+1} \cdot U_i = \left(\frac{1}{2} - j\frac{1}{2}\right) \cdot U_i \\ &= \boxed{U_c = \frac{1}{2} U_i - j\frac{1}{2} U_i} \end{aligned}$$

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Now:

After studying please solve problems
on AC circuit analysis and use with it
Euler's Formula to let concepts sink in.